

Vishnu Priya Nuthanapati

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Summary

Highly motivated graduate in bioinformatics and biomedical engineering, with strong expertise in NGS data processing, Transcriptomics, and omics data analysis. Proficient in various software tools and computational techniques for large-scale data analysis and workflow automation. Experienced in developing efficient data pipelines and conducting comprehensive analyses to extract meaningful biological insights. Seeking a role to apply skills and contribute to innovative scientific advancements in bioinformatics and biotechnology.

Relevant Technical Skills

- R/BioConductor/RShiny
- Docker/Singularity
- Snowflake
- JupyterHub/Notebooks
- SQL/MySQL
- Cloud Services (AWS)
- HPC Computing (Slurm)
- Mage/NextFlow/wdl
- Web Technologies
- Python/BioPython
- Conda/PositConnect
- Rest APIs
- Unix/Shell/Bash
- Pandas/Streamlit/Plotly
- HTML/CSS/NodeJS
- GitHub/BitBucket
- Matlab
- Rshiny

Domains Worked In

- NGS/RNA-Seq (small, bulk, sc)
- Transcriptomics
- Mutation Analysis (Vartrix)
- Benchling
- CITE-Seq
- WGCNA
- Probability & Statistics
- EHR (AllofUs)
- Spatial (Geomx/ VisiumHD)
- Machine/Deep Learning
- Visualization (Cytoscape)
- Pacbio/NanoPore

Work Experience

University of Mississippi Medical Center

Jun 2025 – present

Bioinformatician 1

Mississippi, USA

- Analyzed GeoMx spatial transcriptomics with CellChat to map region-specific expression and signaling, identifying 30+ key ligand-receptor pairs and spatial niches for targeted pathway validation.
- Developed an interactive R Shiny CellAtlas for placental marker exploration, enabling real-time visualization of 20k+ single-cell profiles and reducing marker query and export time by 40%.
- Integrated PacBio methylation, variant, and ClinVar data to prioritize clinically relevant loci, flagging 150+ candidate sites and streamlining patient-centered genomic interpretation by 35%
- Identified alternative splicing changes in genes of interest by quantifying differential exon usage across conditions, uncovering phenotype-associated exon patterns in 10+ key transcripts
- Designed small and bulk RNA-seq pipelines with mixed-effect models, enabling robust differential expression and pathway insights across 200+ samples
- Built de novo transcriptome assemblies for non-model organisms, improving downstream analyses with limited genomic references

Vor Biopharma Inc

Jan 2024 – Jul 2024

Quantitative Biology Co-op

Massachusetts, USA

- Utilized Nextflow pipelines to analyze long-read sequencing data, identifying AML mutations at single-cell resolution with a 15% increase in detection sensitivity
- Leveraged algorithms to track AML blast evolution from diagnosis to relapse, improving mutation tracking accuracy by identifying an additional 20 mutations and providing deeper insights into disease progression
- Developed a data validation pipeline to enrich sequencing data, reducing error rates by five times and enhancing the accuracy of genetic sequencing analysis for more reliable biological insights
- Built ETL pipelines using Mage, reducing data processing time from 12 to 2 hours and improving access to thousands of records daily via REST APIs for biological analysis
- Partnered with the process development team to design, build, and deploy dashboards for biological sample inventory management, achieving a 40% increase in process efficiency and enabling real-time tracking of inventory
- Assessed CRISPR ABE edits from NGS samples and developed visualizations, enabling wet-lab scientists to analyze 200+ samples daily and improving workflow efficiency
- Coordinated with the business development team to develop a comprehensive database of registered sequence listings, streamlining access to over 5,000 records and improving IP management efficiency by 30%

Accenture Private Solutions

Aug 2021 - Jun 2022

Associate Software Engineer

Telangana, India

- Collaborated with the DevOps team to automate DocuSign Access requests using Power BI, processing 50+ requests daily and reducing manual effort
- Led patching on client servers, overseeing scheduling, testing, and deployment of updates for the client operations team
- Streamlined access to GAMS (Global Asset Management System) by managing user accounts with SQL, improving client request response time by 25%, and account management efficiency by 30%
- Supervised clients by providing permissions for information on terminated employees, viewing, and tracking assets globally

Campus Fellowships Mentorship Program

Nov 2020 - Jun 2021

Bioinformatics Research Intern

Telangana, India

- Performed NGS data processing using RSEQREP in AWS, handling large datasets and optimizing pipelines to process thousands of samples faster and improve overall data accuracy for genomic analysis
- Optimized transcriptomic data processing pipelines, including QC (fast QC, multiQC), alignment, and expression quantification, ensuring efficient handling of large-scale datasets and high-quality data for downstream analysis
- Directed downstream genomic data analysis, identifying key regulatory pathways and novel gene targets through differential expression and pathway enrichment
- Designed a mentorship program website with HTML, CSS, NodeJS, and Balsamic, resulting in a 35% increase in student outreach

Knowledge Solutions India

Aug 2020 - Oct 2020

Research Intern

Hyderabad, India

- Developed an insurance premium prediction engine using an ensemble of Multi-Linear and Random Forest Regressors, improving prediction accuracy and aiding in better decision-making
- Implemented feature selection using Principal Component Analysis and achieved $\tilde{1.5}\%$ MAPE on predictions

Gowri Gopal Hospital

Aug 2020 - Oct 2020

BioMedical Intern

Kurnool, India

- Initiated an orientation program on handling basic errors in medical devices with bio-medical professionals for medical interns and doctors across all departments
- Calibrated medical devices and instruments with calibration tools using calibration kits and reduced error rate by 5%

Projects

Mapping of miRNA-Gene regulatory Networks in Schizophrenia Across Tissues

Jul 2021

- Generated comprehensive network map of miRNA-gene interactions in Schizophrenia using the TargetScan prediction framework
- Identified several active slices of this network across multiple tissues using contextual expression-based parameters
- Integrating the tissue-specific network maps, identified genes with maximum convergence across tissues as well as existing FDA-approved drugs targeting them for drug-repositioning

Single-Cell Transcriptomic Profiling of Ribosomal Dysregulation in Tumor-Associated B-Cells

Aug 2024

- Analyzed 17,549 murine lung cells using Seurat, applied Harmony batch correction resolving cohort effects, and identified 13 immune clusters with neutrophil predominance
- Built reproducible scRNAseq pipeline integrating clusterProfiler that improved cluster resolution by 38% and automated visualization of 22 key markers
- Revealed 47 DEGs in tumor B-cells showing ribosomal upregulation and mitochondrial suppression, validated by GO enrichment showing neutrophil activation pathways

Education

Northeastern University

Boston, MA

Master of Science in Bioinformatics; **GPA: 3.9/4.0**

Expected Dec 2024

- **Coursework:** Bioinformatics Computational & Statistical Methods, Advanced Genomics, Omics, ML & AI

Osmania University, College of Engineering

Telangana, India

Bachelor of Engineering in Bio-Medical Engineering; **GPA: 4.0/4.0**

Aug 2017 - Jul 2021

- **Coursework:** Biomedical Instrumentation, Cell-Tissue Engineering, Human Physiology, Biomedical Signal Analysis

Conference & Volunteer Activities

- Presented Poster at the Experimental Biology 2021 Conference
- Volunteered at IEEE, Hyderabad section, as a marketing volunteer (EMBS society)